Mateo Jimenez-Sotelo
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Research Interests

Population genomics, ancient DNA, systems biology, and machine learning for biological data.

Education

National Autonomous University of Mexico (UNAM) Morelos, Mexico
B.Sc. in Genomic Sciences. Aug 2024 - present
GPA: 9.38/10 (≈ 3.9/4.0)

Relevant Coursework: Statistics, Probability, Calculus, Linear Algebra, Differential Equations, Bioinformatics, Genetics, Human & Population Genomics, Transcriptomics, Epigenetics, Molecular & Cell Biology

Research Experience

International Laboratory for Human Genome Research, UNAM Juriquilla, Mexico
Undergraduate researcher Jan 2025 - present
PI: Maria Avila-Arcos, Population & Evolutionary Genomics Lab.

- Analyzed ancient and modern genomic datasets to investigate population structure in colonial-period Mexico
- Performed PCA and admixture analyses integrating ancient individuals from La Conchita and Hospital San José de los Naturales with reference populations from the 1000 Genomes Project and the Mexican Biobank
- Contributed to population genetic interpretation of ancient pathogen research within the “Ancient Pathogens of New Spain” project
- Collaborated with graduate researchers to process genomic datasets and evaluate ancestry components in ancient Mexican populations

Center for Genomics Sciences, UNAM Cuernavaca, Mexico
Undergraduate researcher Dec 2024 - present
PI: Julio A. Freyre-Gonzalez, Systems Biology Lab.

- Contributed to a large-scale metadata curation project integrating RNA-seq and microarray experiments from *Gene Expression Omnibus* (GEO) to build a controlled vocabulary for transcriptomic experiment annotation.
- Investigated natural language processing (NLP) methods for automated classification of transcriptomic experiment metadata.

- Explored fine-tuning strategies for pre-trained large language models (LLMs) using biological text corpora to improve metadata normalization and ontology mapping.
- Completed laboratory training in Python programming, network topology, and systems biology concepts applied to biological data analysis.

Additional Training

Biology Olympiad Training Program - UNAM Mexico City, Mexico
Academic Trainee May 2023 - August 2023
Moises O. Fiesco-Roa, Embryology and Genetics Department, Faculty of Medicine, UNAM.

- Completed interdisciplinary coursework across multiple UNAM institutions, including the Faculties of Medicine and Sciences and the Instituto Nacional de Pediatría (INP).
- Training topics included general biology, genetics, neurophysiology, plant biology, population genetics, mycology, ornithology, developmental biology, and cellular signaling.

Awards

- Gold Medalist, Mexico City Biology Olympiad (OCaBi), 2023

Courses

- **Large Language Models and HF ecosystem** - Hugging Face, Feb 2026
- **Human Population and Evolutionary Genetics** - MOOC, Institut Pasteur, Dec 2025
- **Fundamentals of Deep Learning** - Deep Learning Institute (NVIDIA) and the Institute of Biotechnology (UNAM), Aug 2025.

Skills

Programming Languages: Python (advanced), Bash (advanced), R (intermediate), C (intermediate), C++ (basic), Julia (basic), Perl (basic)
Libraries: Pandas, NumPy, PyTorch, Transformers, SciPy, Scikit-learn, BioPython, pyDESeq2
Bioinformatics: Genome assembly & QC, Metagenomics, Population genomics, Motif analysis, Regulatory Network analysis
Tools: SAMtools, BCFtools, Plink, Admixture, FastQC, SPAdes, Flye, QUAST, PROKKA, Kraken2, RSAT
Computing: Git, HPC environments
Omics: Bulk & single-cell RNA-seq, spatial transcriptomics
Languages: Spanish (native), English (C1)